

Notes & Steps



Key idea

An angle is the amount of turn between two rays from a common vertex. Angles are measured in **degrees ($^{\circ}$)** using a **protractor**.

How to use a protractor

1. Place the centre hole or crosshair exactly on the **vertex** (the corner).
2. Line up the 0° mark with one arm of the angle.
3. Read the number where the other arm crosses the scale.
4. Check which scale (inner or outer) starts at 0° and use that one.

Key idea

Angle types:

Protractor tips

- ▶ Always start from 0° , not from the edge of the protractor
- ▶ If the angle opens to the left, use the outer scale; if to the right, use the inner scale (or vice versa — check which one starts at 0)
- ▶ For accuracy, estimate the angle first — if it looks like 30° , don't read 150°
- ▶ The two scales add up to 180° (inner + outer = 180°)

Common mistake

Reading the wrong scale. Estimate the angle type first. If the angle is acute (looks small), the answer should be less than 90° . If you read 120° on an acute-

Notes & Steps



Estimating angles

Before measuring, practise estimating:

- ▶ A full turn = 360°
- ▶ Half a turn = 180° (straight line)
- ▶ Quarter turn = 90° (right angle)
- ▶ A slightly open door $\approx 30^\circ$
- ▶ A wide open door $\approx 120^\circ$

Example 1

Estimate then measure: the angle looks about half a right angle ($\approx 45^\circ$). When measured, it reads 42° — close!

Example 2

Draw a 60° angle: draw a baseline ray, place protractor at endpoint, mark at 60° ,

Try these

1. Estimate: what type of angle is 85° ?
2. Estimate: what type of angle is 195° ?
3. If the inner scale reads 40° , what does the outer scale read?
4. A student reads 130° for an acute-looking angle. What went wrong?

Common mistake

Not placing the protractor centre on the vertex. Even being 1 mm off can cause a $2\text{--}3^\circ$ error. The measurement must start from the exact vertex.